

Ahmed Fathy

Software Engineer

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Experience

Al Nada Scientific Office / IT Specialist

June 2022 - 2024, Cairo, Egypt

- **Database Application Development:** Designed and upgraded a database application from VBA and Microsoft SQL to React, Express.js, and PostgreSQL for improved scalability.
- **Technical Maintenance:** Hands-on experience in maintaining and troubleshooting electronic devices.
- **IT Systems Support:** Collaborated with a team to implement and support IT systems efficiently.

Education

- **Cairo University Faculty of Engineering / B.Sc. in Computer Engineering** 2023 - 2027
GPA: 3.6

Projects

Tayseer-Web([Preview](#))([Github](#)):

Built a corporate website for a company using Next.js and TypeScript.

- Designed and developed responsive pages introducing the company, services, and contact details.
- Leveraged Next.js's server-side rendering (SSR) and TypeScript's static typing for performance and reliability.
- Styled with Tailwind CSS for a clean and professional look.

Company Manager App (Al Nada Scientific Office) : using React.js, Redux, Express.js, Prisma, and PostgreSQL.([Github](#))

- Implemented CRUD operations for customers, suppliers, stock, sales, purchases, debts, and price quotations.
- Built a modular backend with Prisma ORM and RESTful APIs for efficient data handling.
- Styled UI with Tailwind CSS and integrated Redux for state management.

Blog Web Application : Built a full-stack blog platform using Node.js, Express.js, EJS, and PostgreSQL.([Preview](#))([Github](#))

- Implemented CRUD operations, styled content via Quill editor, and image uploads with Multer.
- Designed modular backend architecture with clean routing and middleware layers.

Interactive Orrery Web Application : Created a 3D solar system visualizer using React.js and Three.js.([Preview](#))([Github](#))

- Rendered real-time planetary motion and near-Earth object data with responsive controls.

Tactic (تكتيك) : Developed a browser-controlled game suite integrating AI, computer vision, and robotics.([Github](#))

- Used FastAPI, YOLOv5, and OpenCV for backend logic and object detection.
- Built the frontend with Next.js and connected to ESP32 hardware via WebSockets.

Alien Invasion Game (C++) : Designed a simulation game modeling combat scenarios using custom data structures.

Paint for Kids Game (C++) : Built an OOP-based drawing application for children with interactive UI and basic tools.

Path Planning Algorithm Study ([Github](#)): Compared A*, Dijkstra's, and RRT algorithms through C++ implementations and performance analysis.

Courses

[The Complete Web Development Bootcamp](#) - Udemy (2023)

Key topics: HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Git, REST APIs, and deployment.

[The Ultimate React Course: React, Redux & More](#) - Udemy (2025)

Key topics: React fundamentals, hooks, Redux, routing, context API, performance optimization, and Next.js.

Skills

Tools & Technologies

- **Languages:** JavaScript, Python, C++, SQL
- **Frontend:** React, Next.js, EJS, Tailwind CSS, Material-UI
- **Backend:** Node.js, Express, REST APIs
- **Databases:** MongoDB, PostgreSQL
- **DevOps & Tools:** Docker, Docker Compose, Git, GitHub
- **Concepts:** Data Structures & Algorithms